

Qatar-10b

R-0079

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_241022 · created 2026-07-15T20:19:12+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460605.6413 +/- 0.0088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.04
Transit depth (Rp/Rs)^2	2.59 +/- 1.28 [%]
Orbital Inclination (inc)	85.78 +/- 2.53
Airmass coefficient 1 (a1)	1.394 +/- 0.021
Airmass coefficient 2 (a2)	-0.182 +/- 0.029
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	2.68
Optimal Annulus	7.22
Transit Duration (day)	0.037 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.161 $\pm$ 0.04 vs 0.1265 (deviation 0.86 σ)
Duration fit vs published	0.037 d vs 0.1154 d (deviation 7.84 σ)
Mid-time O-C	-27.3 min
Depth SNR	4.0
Residual scatter	1.47 %

Light curve

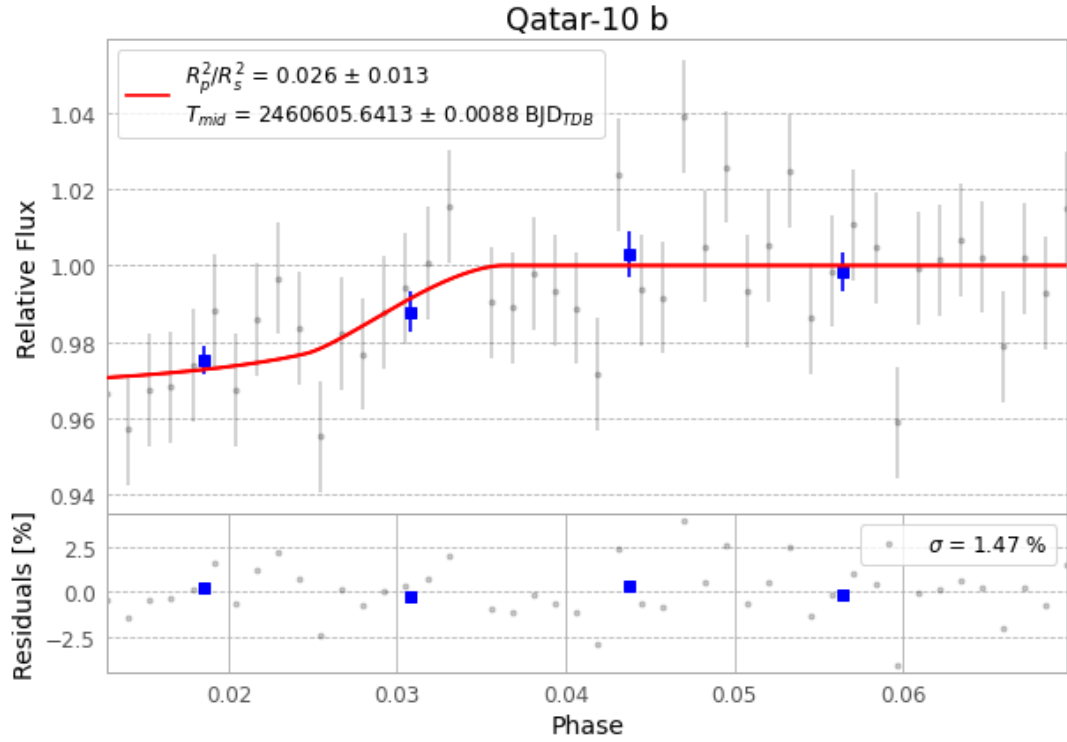


Figure 1 — FinalLightCurve_Qatar-10 b_2024-10-21.png (SHA-256 1d14a7d9a55dd458...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 294], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5969d9d5e5486270...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

46 FITS frames (30 MB), Qatar-10241022035115.FITS → Qatar-10241022060615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-241022.log (262732dd971b...), FinalLightCurve_Qatar-10 b_2024-10-21.png (1d14a7d9a55d...), FinalLightCurve_Qatar-10 b_2024-10-21.csv (3a4de5876aba...)

Compute resources

Wall time 100.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 20:19Z.